Parul® University
Vadodara, Gujarat
NAAC GRADE A++

Information and
Communication Technology

Unit 1 : Introduction to AI Programming

Artificial Intelligence & Data Science

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Content

Introduction

➢ Definition:

Artificial Intelligence (AI) refers to the simulation of human intelligence processes by computer systems. These processes include learning, reasoning, problem-solving, perception, and language understanding.

Historical Context

First coined by John McCarthy in 1956 at the Dartmouth Summer Research Project. AI has evolved through multiple "winters" and "springs," with modern deep learning breakthroughs from 2012 onwards.

What is AI and Machine Learning?

- **Artificial Intelligence (AI):** Broad field of computer science focused on creating intelligent systems
- **Machine Learning (ML):** A subset of AI enabling systems to learn from data without being explicitly programmed.
- **Deep Learning (DL):** A subset of ML using neural networks with multiple layers for complex pattern recognition.

Relation Between AI,ML and DL

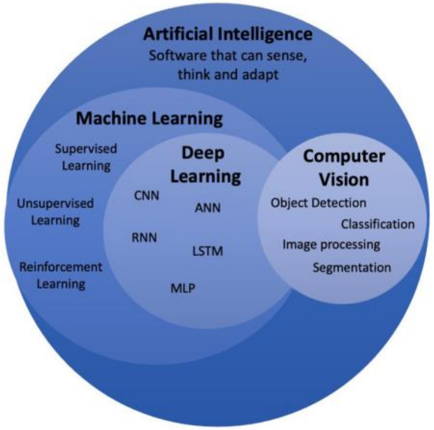

Figure 1.1 Op-Amp Symbol

Real-World AI Applications

1. **Healthcare:** Disease diagnosis, medical imaging analysis, drug discovery and personalized medicine.

2. **Autonomous Vehicles:** Self-driving cars using computer vision and intelligent decision-making systems.

3. **Natural Language Processing:** Chatbots, machine translation, sentiment analysis, text summarization.

4. **Recommendation Systems:** Netflix, Amazon, Spotify personalized recommendations using collaborative filtering.

5. **Financial Services:** Fraud detection, algorithmic trading, credit risk assessment, portfolio optimization

Why Python for AI Programming?

Ease of Learning: Simple, readable syntax enabling rapid development and quick prototyping

Rich Libraries: NumPy, Pandas, TensorFlow, scikit-learn, PyTorch, Keras for all AI/ML needs

Large Community: Extensive documentation, tutorials, and active community support worldwide

Performance: Fast execution for numerical and data operations with C/C++ backend optimizations.

Industry Standard: Widely adopted in academia, industry, and cutting-edge AI research globally

Python Fundamentals for AI

Essential Python Concepts:

- Variables, data types (int, float, string, list, dict, tuple, set)
- Control flow (if-else statements, loops, functions, comprehensions)
- Object-oriented programming (classes, objects, inheritance, polymorphism)
- List and dictionary comprehensions, lambda functions, map/filter
- Exception handling, debugging, and code optimization techniques

Essential AI Libraries

NumPy - Numerical computing, arrays, matrices, mathematical operations
Pandas - Data manipulation, analysis, DataFrames, CSV/Excel file handling
Matplotlib - Data visualization, plotting, statistical graphics, image display
scikit-learn - Machine learning algorithms, classification, regression, clustering
TensorFlow/PyTorch - Deep learning frameworks, neural networks, GPU acceleration

Best Practices in AI Programming

Code Organization: Modular code structure, clear function/variable names, comprehensive documentation

Version Control: Use Git for tracking code changes, collaboration, and project management

Testing: Unit testing, integration testing, validation of edge cases and results

Code Style: Follow PEP 8 guidelines, use linters, maintain consistent formatting throughout

Documentation: Explain logic, assumptions, algorithms, parameters, and return values clearly.

AI Development Environments

Three popular environments for AI programming:

1. **Jupyter Notebook:** Interactive, web-based notebooks for experimentation and data exploration
2. **Google Colab:** Free cloud-based Jupyter environment with free GPU/TPU support and automatic saving.
3. **VS Code:** Professional IDE with powerful extensions for AI development and production code

Jupyter Notebook Setup

Installation Steps

- Install Python 3.7 or later from python.org
- Run command:
- Launch Jupyter:
- Browser opens with Jupyter interface at localhost:8888Key Activities:

Features: Mix code, output, and markdown in cells; visualization support; easy sharing

Google Colab Setup and Usage

Getting Started

- Visit **colab.research.google.com**
- Sign in with your Google account
- Create new notebook or upload existing file
- Access free GPU/TPU for accelerated computation
- Automatic saving to Google Drive.

Advantages: Zero setup, cloud-based, pre-installed libraries, free GPUs, easy collaboration

VS Code Configuration for AI

Essential Extensions

- **Python Extension (Microsoft):** IntelliSense, debugging, linting capabilities
- **Pylance:** Advanced static analysis, type checking, code intelligence
- **Jupyter Extension:** Run Jupyter notebooks directly within VS Code
- **GitLens:** Git integration, version control, collaboration features
- **Better Comments:** Highlight and organize important code sections

VS Code Setup Steps

- Download and install VS Code from code.visualstudio.com
- Install Python 3.7+ from python.org
- Open VS Code Extensions marketplace (Ctrl+Shift+X)
- Search and install "Python" extension by Microsoft
- Select Python interpreter (Ctrl+Shift+P → Select Interpreter)
- Create virtual environment for your project
- Install required AI libraries

Comparison of Development Environments

Setup	Local Installed Required	No setup Nedded	Local Installed Required
GPU ACCESS	Local GPU Required	Free GPU Available	Local GPU Required
Best for	Experimentation	Learning and prototyping	Production Code
Collaberation	Manual Sharing	Easy Sharing Via Link	Requires Git

Your First AI Program

Simple Example: Predicting House Prices with Linear Regression

```
from sklearn.linear_model import LinearRegression

# Data (size → price)
X = [[500], [700], [900], [1100]]  # sq ft
y = [25, 35, 45, 55]               # price in lakhs

# Train model
m = LinearRegression().fit(X, y)

# Predict
print(m.predict([[800]])[0])  # price for 800 sq ft
```

References

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- **Chollet, F. (2021).** *Deep Learning with Python.* Manning Publications
- **TensorFlow Official Documentation:** www.tensorflow.org
- **PyTorch Official Documentation:** www.pytorch.org
- **Scikit-learn Documentation:** **www.scikit-learn.org**

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